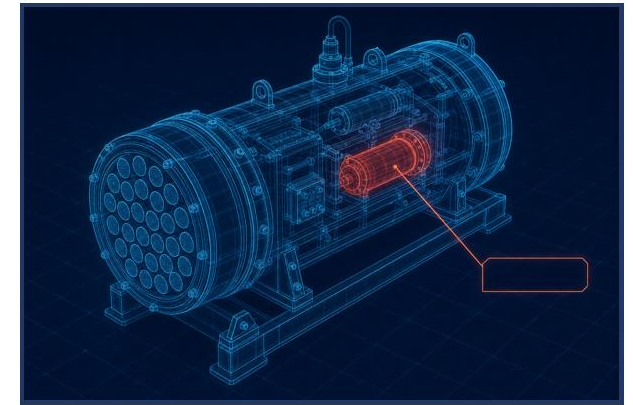


TITLE PAGE

SMART INDIA HACKATHON 2026

Problem Statement ID	26075
Problem Statement Title	CAPACITY CONNECT — A Digital Capacity Building and Learning Management Portal
Theme	Smart Education
PS Category	Software
Team ID	Team ID
Team Name (Registered on portal)	Team Name



Digital Twin concept (illustrative)

ORGANISATION

**Ministry of Earth Sciences
(MoES)**

DEPARTMENT

**India Meteorological
Department**

One portal for training, assessment and competency — that keeps working when the network does not.

IDEA TITLE

CAPACITY CONNECT — one offline-first portal for learning, assessment, competency mapping and AI-assisted field troubleshooting

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

- **Single portal, three roles. Trainee, Trainer and Admin** — secure signup/login with Admin approval and role management.
- **Trainee: professional profile (qualifications, experience, skills, interests, certificates), course enrollment, learning resources, subject-wise MCQ assessments and course feedback.**
- **Trainer: competency declaration, questionnaires with deadlines, participation and performance monitoring, and a Trainer Library of recorded lectures, presentations and study material.**
- **Admin: user approval, role management, dashboards for courses / enrollments / certifications / assessments / participation, plus homepage publishing of notifications, announcements and achievements.**
- **Competency mapping engine scores trainers against subject requirements to objectively identify the right trainer.**
- **Offline-first edge app: a real local database plus a local AI node on 127.0.0.1, so work continues at sea and at remote sites.**

How it addresses the problem

- Replaces scattered drives, mail and spreadsheets with one auditable source of truth for learning and competency.
- Removes the connectivity barrier: every feature works offline and reconciles automatically on reconnect.
- Turns tacit instrument know-how into retrievable, guided diagnostic procedures.



Concept UI mockup — offline dashboard and mobile learning app

Innovation and uniqueness of the solution

Offline is the default, not a fallback — the local DB is an operating database, not a cache

Local AI engine, no cloud LLM — deterministic retrieval pipeline on 127.0.0.1

Learning fused with operations — a fault maps to the asset and to the training that covers it

3D Digital Twin guidance — SONAR-001 → Mesh_042 shown CRITICAL from a cached GLB

Objective competency mapping — fit computed from skills, certificates and outcomes

Auditable sync ledger — every offline action is versioned and conflict-checked

TECHNICAL APPROACH

Technologies to be used (programming languages, frameworks, hardware)

CLIENT

- Flutter 3 / Dart, Material 3
- Riverpod state · GoRouter
- Drift + SQLite (SQLCipher)
- flutter_secure_storage · Dio
- Android · Windows · Linux · Web

LOCAL AI

- **Python 3.11 · FastAPI on 127.0.0.1**
- **Normalize · language detect · tokenize**
- **Keyword / phrase / RapidFuzz match**
- TF-IDF knowledge retrieval
- Pluggable local LLM slot

3D & MEDIA

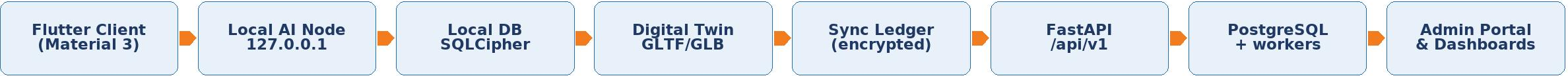
- GLTF / GLB cached locally
- **Mesh mapping + fault state layer**
- Chunked resumable downloads
- Streaming video · PDF viewer
- Optional speech-to-text input

BACKEND & OPS

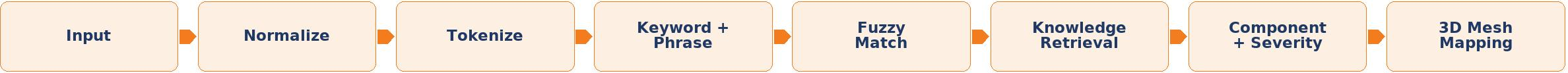
- FastAPI · SQLAlchemy · Pydantic
- **PostgreSQL + Alembic migrations**
- Celery workers · JWT + refresh
- Docker · GitHub Actions CI
- pytest · flutter test/analyze

Methodology and process for implementation

A. System architecture — edge application works with zero connectivity; cloud platform is independently deployable



B. Local AI troubleshooting pipeline — returns structured JSON, confidence is a defined weighted score



Worked example: "Sonar transducer showing abnormal vibration and casing fracture" → SONAR-001 · Sonar Transducer Array · Mesh_042 · severity HIGH · confidence 0.94 → Digital Twin highlight + diagnostic procedure + diagnostic record saved offline.

FEASIBILITY AND VIABILITY

Analysis of the feasibility of the idea

Technically proven stack

Flutter, FastAPI and PostgreSQL are mature and fully open-source; one Dart codebase covers Android, Windows, Linux and Web.

Zero licensing cost

Entirely open-source; no per-seat fees, so scaling to more users and institutes costs only storage and compute.

No exotic hardware

Runs on existing MoES/IMD field tablets and vessel workstations; backend is containerised on existing infrastructure.

Incremental rollout

Portal features usable from day one; AI and Digital Twin layers extend the same schema without rework.

Potential challenges and risks → Strategies for overcoming these challenges

CHALLENGE / RISK	MITIGATION STRATEGY
Poor / no connectivity at sea and remote sites	Offline-first architecture: local operating DB + local AI node; sync ledger drains automatically on reconnect.
Large 3D models on low-end field devices	Decimated LOD meshes by default, full-resolution GLB as optional download; renderer degrades to metadata-only view.
Offline credential caching weakens authentication	Admin-approved devices only, Argon2id verifier (never the password), bounded grace window, audit + re-verification on sync.
Sync conflicts on shared records	Per-entity strategies (version compare, field merge, manual queue); diagnostic records are never silently overwritten.
Digital literacy and user adoption	Role-tailored simple UI, localisation-ready strings for Indian languages, optional voice input, in-app guided demo scenario.
Storage exhaustion on field devices	Storage manager with per-category usage, resumable downloads, permissioned removal of optional assets only.

IMPACT AND BENEFITS

Potential impact on the target audience

TRAINEES / FIELD ENGINEERS

- Learn, test and certify with zero connectivity – at sea and at remote sites
- Guided AI diagnostics instead of waiting for shore support
- Portable, verifiable competency profile

TRAINERS

- One versioned library for lectures, presentations and material
- Questionnaires with deadlines, auto-scoring, live participation view
- Feedback shows which content actually works

ADMINISTRATION / MoES

- Live dashboards of enrollment, certification and assessment
- Objective trainer selection via competency mapping
- Complete audit trail for governance and reporting

Benefits of the solution (social, economic, environmental)

SOCIAL

Equitable access to training for remote and shipboard staff; skill recognition through verifiable certificates; localisation-ready for Indian-language expansion; safer field work through guided procedures.

ECONOMIC

Lower instrument downtime and fewer avoidable failures; reduced travel and physical-classroom cost; open-source stack with no licensing fees; reusable across INCOIS, NCPOR and NIOT on the same core.

ENVIRONMENTAL

Fewer service trips and paper-free assessments; longer instrument life through timely preventive maintenance; better-maintained ocean and weather sensors mean higher-quality earth-science data.

100%

core features usable offline

↓ 40%

target fault-resolution time

3

roles, one governed platform

4

platforms, one codebase

RESEARCH AND REFERENCES

Details / Links of the reference and research work

Problem & organisational context

- **SIH 2026 Problem Statement 26075 — Ministry of Earth Sciences (MoES) — sih.gov.in**
 - [maes.gov.in](https://mausam.imd.gov.in) | India Meteorological Department — mausam.imd.gov.in
- MoES Annual Report — capacity building & human-resource development
- National Education Policy 2020 — digital, competency-based learning
- SIH 2026 portal guidelines — idea submission format and evaluation criteria

Standards & frameworks referenced

- SCORM / xAPI (Experience API) — learning-record interoperability
- IEEE 1484 LOM — learning-object metadata for the course catalog
- NIST SP 800-63B — digital identity & authentication guidance
- OWASP ASVS + Mobile Top 10 — application security verification baseline
- NIST SP 800-38D — AES-256-GCM authenticated encryption
- ISO/IEC 27001 Annex A — access control, logging, audit-trail controls

Technology documentation

- Flutter & Dart — docs.flutter.dev | Riverpod, GoRouter, Drift
- FastAPI — fastapi.tiangolo.com | SQLAlchemy, Alembic, Pydantic official docs
- SQLCipher — full-database encryption for SQLite (zetetic.net/sqlcipher)
- Khronos glTF 2.0 specification — 3D asset format for the Digital Twin
- PostgreSQL 16 docs | Docker Compose deployment reference

Domain & prior art studied

- Digital Twin concepts for marine instrumentation (ISO 23247 framing)
- Argo programme documentation — argo.ucsd.edu — float operations
- Offline-first patterns; CRDT vs. ledger-based synchronisation literature
- Open-source LMS study (Moodle, Open edX) — role model & gap analysis
- RapidFuzz / TF-IDF retrieval literature — basis of the confidence score

Note: all cited standards and libraries are open or publicly available; the solution uses no proprietary or licence-restricted component.